Individual Task - Module 3

Machine Learning: Concepts, Algorithms, and Applications

1)Bayes' theorem in real life: Email spam Detection.

Introduction

In today's digital world, email has become one of the most important modes of communication.

However, along with useful emails, users also receive a large number of unwanted emails called spam.

These spam emails may contain advertisements, fake offers and even harmful links.

To protect users, email service providers use machine learning algorithms to automatically classify emails as spam or not spam.

One of the important mathematical concepts used in this process is Bayes' Theorem.

Bayes' Theorem helps in calculating the probability of an event based on prior knowledge and new evidence.

In email spam detection, Bayes' Theorem is used to find the probability that an email is spam based on the words and features present in it.

Bayes' Theorem plays a vital role in machine learning applications like email spam detection. By using prior probabilities and new evidence, it helps classify emails accurately.

Bayes' Theorem - Definition.

Bayes theorem is a principle in probability theory that describes how to update the probability of a hypothesis when new evidence is available.

Mathematical Formula:

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

Where:

P(A|B) = Probability of event A given B.

P(B|A) = Probability of event B given A.

P(A) = Probability of event A.

P(B) = Probability of event B.

Understanding Bayes Theorem in Email Spam Detection

In email spam detection:

- A = email is SPAM
- B = Email contains certain words like "free", "win", "offer".

Baye's theorem helps us calculate:

"What is the probability that an email is spam, given that it contains certain spam-related words?"

Real-Life Example with Calculation

Let us consider a simple example.

Assume:

30% of all emails are spam ->P(Spam)=0.3

70% of emails are not spam->P(Not Spam)=0.7

80% of spam emails contain the word "free" -> P (Free | spam) = 0.8.

10% of non-spam emails contain the word "free" -> P (Free \ not spam) = 0.1

step 1:

Calculate P(Free)

P(Free) = (0.8 x 0.3) + (0.1 x 0.7)

P(Free) = 0.24 + 0.07 = 0.31

step 2:

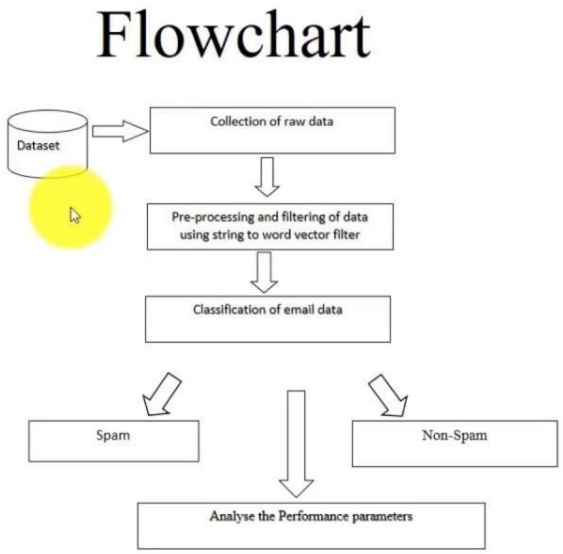
Apply Bayes' theorem

P(spam | Free) = 0.8X0.3/0.31

P(Spam | Free) =0.774

Result: There is a 77.4% probability that an email containing the word "free" is spam.

“The diagram shows how Bayes' theorem is used to classify emails as spam or non-spam based on Probability.”



Importance of Bayes' theorem in machine learning

Helps in probability-Based Decision making:

- machine learning system often need to make decision based on probabilities rather than exact values.
- Bayes' theorem allows models to update their prediction when new data becomes available.

Used in real-world Applications ○ Bayes' theorem is used in many real-world machine learning applications.

○ In spam filtering, it helps identify unwanted emails.

works well even with limited data: ○ one of the major advantages of Bayes' theorem is that it works effectively even when only a small amount of data is available.